Group II, Row IV: subtracting both poles ($\xi \to 0$ and $\xi \to 1$) from the start

Both rapidity-sensitive regions are removed at the level of the master expression, with the z-integral still undone; every counterterm is subtracted once and added back once, so no regrouping or relocation of terms is needed afterwards. The sequence is: master → per-ξ form → two endpoint subtractions → (JIMWLK + endpoint) and (BFKL + boundary) from the counterterms → z-integration of the doubly subtracted remainder → DGLAP + RC + azimuthal.

1 From the master to the per-ξ form (no z-integral yet)

The starting point is the master expression

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'} e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m\,(x-y)^k\,(w'-z)^{m'}\,(x'-y')^{k'}}{(w-z)^2\,(x-y)^2\,(w'-z)^2\,(x'-y')^2}\,\frac{p^+\,k^+}{(p^++k^+)^4}\\ \delta^{(2)}\left[y-w+\frac{p^+}{(p^++k^+)}\,(w-z)\right]\times\,\delta^{(2)}\left[y'-w'+\frac{p^+}{(p^++k^+)}\,(w'-z)\right]\times\\ \left[\delta_{km}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\times\left[\delta_{k'm'}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]\\ \left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right]\end{aligned}\tag{1}$$

(obtained from the operator setup). Perform, exactly as before: the colour algebra ($U^{d'b}(z)U^{db}(z)=\delta^{d'd}$, $f^{c'da}f^{cda}=N_c\delta^{c'c}$, giving $N_c\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\,\rho^{e}(x)$); the δ-function integration over w,w' (Jacobian $\left[\frac{p^++k^+}{k^+}\right]^2$ per leg, $w-z=\frac{p^++k^+}{p^+}(y-z)\,\ldots$ equivalently $y-z=\frac{p^+}{p^++k^+}(w-z)$, and phase $e^{-ik(y'-y)\frac{p^++k^+}{k^+}}$); the contraction of the two tensor brackets over i,j; and the substitution $\xi=\frac{k^+}{p^++k^+}$, $dp^+=-\frac{k^+d\xi}{\xi^2}$ (the descending limits absorb the sign). The antisymmetric $\frac{2}{k^+}\left(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km}\right)$ channel is dropped at once: for every term whose colour factor is z-independent, the z-integral $\int_z\mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)$ is symmetric in (m,m'), so the antisymmetric contraction vanishes identically at each ξ. The result, with the *z-integral still undone*, is

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{\bar\xi}^{\xi_{\max}}\frac{d\xi}{\xi^2}\int_{x,x',y,y',z} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\Big\{\,d_\perp\,\xi(1-\xi)\left[\mathcal{K}(y-z)\cdot\mathcal{K}(x-y)\right]\left[\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-y')\right]\\ +\left[\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\right]K(y,y',z)\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\Big\}\,\left[U^{ce'}(y')-U^{ce'}(z\right.\end{aligned}\tag{2}$$

with $\bar\xi=\frac{k^+}{\vee}$, $\xi_{\max}=\frac{k^+}{k^++\Lambda}$, $\mathcal{K}^m(u)=\frac{u^m}{u^2}$, $K(a,b,z)=\mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$. (Check: carrying out $\int_z\mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)=-\frac{\pi}{2}L_\epsilon\delta^{mm'}-\pi\frac{r^mr^{m'}}{r^2}$, $r=y'-y$, on (2) reproduces verbatim the $d_\perp$-form used previously, so (2) is the same cross-section, one step earlier.)

2 The two poles and their counterterms

The curly bracket of (2) is singular at the two ends of the ξ range, and nowhere else:

- $\xi\to 1$ ($p^+\to 0$): only the channel $\frac{\xi}{1-\xi}$ blows up. The unmeasured gluon becomes soft on the *target* side — this is the JIMWLK region, regulated by Λ through $\xi_{\max}$. At the endpoint the integrand is smooth in the positions: the phase $\to e^{-ik\cdot(y'-y)}$ and $\frac{1}{\bar\xi}\to 1$.
- $\xi\to 0$ ($p^+\to\vee$): the channel $\frac{1-\xi}{\xi}$, together with the measure $\frac{d\xi}{\xi^2}$ and the phase, hides a logarithm of $\frac{\vee}{k^+}$. The unmeasured gluon carries almost all of the *projectile* momentum — the BFKL region. In the positions this is the collapse $y,y'\to z$ (from the δ-functions of (1), $y-z=\xi(w-z)$, $y'-z=\xi(w'-z)$ at fixed w,w').

Define one counterterm per pole, each equal to the exact limit of the corresponding channel:

$$T_J(\xi) = \frac{1}{1-\xi}\, e^{-ik\cdot(y'-y)}\, K(y,y',z)\, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, \Big[U^{ce'}(y') - U^{ce'}(x')\Big]\, [U^{ce}(y) - U^{ce}(x)]\, \rho^{e'}(x')\, \rho^{e}(x)\,, \tag{3}$$

$$T_B(\xi) = \frac{1}{\xi^2}\,\frac{1-\xi}{\xi}\, e^{-ik\cdot(y'-y)\frac{1}{\xi}}\, K(y,y',z)\, K(x,x',z)\, \Big[U^{ce'}(z) - U^{ce'}(x')\Big]\, [U^{ce}(z) - U^{ce}(x)]\, \rho^{e'}(x')\, \rho^{e}(x)\,. \tag{4}$$

T_J is the $\frac{\xi}{1-\xi}$ channel with every regular factor set to its $\xi = 1$ value (the full integrand there is $\frac{1}{\xi(1-\xi)} e^{-ik(y'-y)/\xi} K\, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, W$, so the difference carries $\frac{e^{-ik(y'-y)/\xi}/\xi - e^{-ik(y'-y)}}{1-\xi}$, finite at $\xi = 1$). T_B is the $\frac{1-\xi}{\xi}$ channel with the slow factors frozen at the collapse point, $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, W(y,y') \to K(x,x',z)\, W(z,z)$; their difference is the single bracket

$$\frac{1}{\xi^2}\,\frac{1-\xi}{\xi}\, e^{-ik\cdot(y'-y)\frac{1}{\xi}}\, K(y,y',z)\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, W(y,y') - K(x,x',z)\, W(z,z)\Big]\,, \tag{5}$$

whose square bracket vanishes linearly in ξ at fixed (w,w',z), because $\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) = K(x,x',z)$ identically. With $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5}\frac{1}{k^+}$ the exact, one-line decomposition of (2) is

$$\boxed{\frac{d^3N_{\rm NLO}}{d^3k} = \underbrace{\mathcal{N}\int d\xi \int \Big[\,\{\text{integrand of (2)}\} - T_J(\xi) - T_B(\xi)\Big]}_{d^3N_{\rm reg}:\ \text{finite at both endpoints}} + \underbrace{\mathcal{N}\int d\xi \int T_J}_{J} + \underbrace{\mathcal{N}\int d\xi \int T_B}_{B}} \tag{6}$$

— each counterterm subtracted once and added back once; nothing is moved twice. In $d^3N_{\rm reg}$ the limits can be taken to $[\bar\xi, 1]$ at a cost $O(\Lambda)$, and the $\bar\xi$ end produces no large logarithm.

3 The $\xi \to 1$ counterterm: JIMWLK + endpoint term

T_J depends on ξ only through $\frac{1}{1-\xi}$, so

$$J = \mathcal{N}\Big[\int_{\bar\xi}^{\xi_{\max}} \frac{d\xi}{1-\xi}\Big] \int_{x,x',y,y',z} e^{-ik\cdot(y'-y)}\, K(y,y',z)\, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, \Big[U^{ce'}(y') - U^{ce'}(x')\Big]\, [U^{ce}(y) - U^{ce}(x)]\, \rho^{e'}(x')\, \rho^{e}(x)\,, \tag{7}$$

The $\log\frac{k^+}{\Lambda}$ part is the **JIMWLK term**, already in its manifestly-μ-free z-form,

$$\boxed{\frac{d^3N_{\rm JIMWLK}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',y'} e^{-ik\cdot(y'-y)}\, K(y,y',z)\, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, \Big[U^{ce'}(y') - U^{ce'}(x')\Big]\, [U^{ce}(y) - U^{ce}(x)}$$

(8)

(equivalently, with $\frac{1}{\pi}\int_z K(y,y',z) = -\frac{2}{\epsilon} - 2\gamma_E - \log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4} \equiv -L_\epsilon$, the L_ϵ-form with the all-negative bracket). The $\log(1-\frac{k^+}{\vee})$ part is the **endpoint (plus-prescription convention) term**; z-integrated it reads

$$-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\, L_\epsilon\, \Big[U^{ce'}(y') - U^{ce'}(x')\Big]\, [U^{ce}(y) - U^{ce}(x)]\, \rho^{e'}(x')\, \rho^{e}(x)\,. \tag{9}$$

4 The $\xi \to 0$ counterterm: BFKL + boundary term

T_B depends on ξ only through the phase and explicit powers. The exact substitution $y = z + \xi(w-z)$, $y' = z + \xi(w'-z)$ gives $d^2y\, d^2y' = \xi^4 d^2w\, d^2w'$, $K(y,y',z) = K(w,w',z)/\xi^2$ and $e^{-ik(y'-y)/\xi} = e^{-ik(w'-w)}$ exactly, so the transverse integral of T_B is $\xi^2 \times$ (ξ-independent), the ξ^2 cancels the measure, and

$$B = \mathcal{N}\Big[\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\Big]\,\mathcal{I} = \mathcal{N}\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]\mathcal{I}\,, \qquad \mathcal{I} = \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)} K(w,w',z)\, K(x,x',z)\, \Big[U^{ce'}(z) - U^{ce'}(x')\Big]\, [U^{c} \tag{10}$$

The $\log\frac{\vee}{k^+}$ part is the **BFKL term**,

$$\boxed{\frac{d^3N_{\rm BFKL}}{d^3k} = +\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\frac{\vee}{k^+}\,\mathcal{I}} \tag{11}$$

(UV finite, μ-independent), and the remainder is the **boundary term** $-\mathcal{N}\,(1-\frac{k^+}{\vee})\,\mathcal{I}$.

5 The doubly subtracted remainder: DGLAP + RC + azimuthal

$d^3N_{\rm reg}$ contains three structures: the $d_\perp\xi(1-\xi)$ channel, the locally subtracted soft channel $\frac{\xi}{(1-\xi)_+}$ (the $\frac{\xi}{1-\xi}$ channel minus T_J), and the μ-free pair (the $\frac{1-\xi}{\xi}$ channel minus T_B, eq. (5)). The pair is finite in $d=2$ and is *kept as it is*. In the other two structures the colour factor is z-independent, so the z-integral is carried out in dimensional regularization:

$$\int_z \mathcal{K}^m(y-z)\,\mathcal{K}^{m'}(y'-z) = -\frac{\pi}{2}\,L_\epsilon\,\delta^{mm'} - \pi\,\frac{r^m r^{m'}}{r^2}\,, \qquad r = y'-y\,. \tag{12}$$

Channel by channel (prefactor $\mathcal{N}$, measure $\frac{d\xi}{\xi^2}$):

(a) $d_\perp\,\xi(1-\xi)\times\delta^{mm'}$ part: $\frac{d_\perp}{2}L_\epsilon = L_\epsilon - 1 + O(\epsilon)$, so it yields $-\xi(1-\xi)\,(L_\epsilon-1)$ on $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')$;

(b) $d_\perp\,\xi(1-\xi)\times\frac{r^m r^{m'}}{r^2}$ part: the r-tensor (azimuthal) structure, coefficient $-d_\perp\,\xi(1-\xi)$;

(c) $\frac{\xi}{(1-\xi)_+}$ channel: $\int_z K = -\pi L_\epsilon$ exactly (the $d_\perp$-trace and the $\frac{rr}{r^2}$-trace combine into L_ϵ with no leftover), giving $-\frac{\xi}{(1-\xi)_+}\,L_\epsilon$ on $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')$.

Using $\frac{\xi(1-\xi)}{\xi^2} = \frac{1-\xi}{\xi}$, the "$-1$" of (a) and the r-tensor of (b) combine into the azimuthal term, and the L_ϵ pieces of (a) and (c) assemble into the soft-channel-removed splitting function $\mathcal{P}_{gg,+}(\xi) - 2N_c\frac{1-\xi}{\xi} = 2N_c\big[\xi(1-\xi) + \frac{\xi}{(1-\xi)_+}\big]$. Finally adding and subtracting $\frac{\beta_0}{2}\delta(1-\xi)$ ($\beta_0 = \frac{11N_c}{3}$; the δ sits at $\xi=1$, where $\frac{1}{\xi^2}=1$ and the phase is the LO phase) splits off the running-coupling term:

$$\begin{aligned}
\frac{d^3N_{\rm reg}}{d^3k} =& \underbrace{-\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2}{8\pi^2}\,\frac{1}{k^+}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi) - 2N_c\frac{1-\xi}{\xi}\Big]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\;L_\epsilon\;\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\,[U^{ce}(y}_{\text{DGLAP (non-soft)}} \\
&\underbrace{+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\,K(y,y',z)\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') - K(x,x',z)\,W(z,z)\Big]\rho\rho}_{\text{DGLAP (soft pair; free of }\frac{2}{\epsilon},\ \mu\text{ and }\log\frac{\vee}{k^+})} \\
&\underbrace{+\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2\beta_0}{16\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\;L_\epsilon\;\Big[U^{ce'}(y') - U^{ce'}(x')\Big]\,[U^{ce}(y) - U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)}_{\text{RC}} \\
&\underbrace{+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') - 2\,\frac{(\mathcal{K}(x-y)\cdot r)(\mathcal{K}(x'-y')\cdot r)}{r^2}\Big]\,\Big[U^{ce'}(y') - U^{ce'}(}_{\text{azimuthal}}
\end{aligned} \tag{13}$$

(in the soft-pair line the colour brackets are those written in (5), abbreviated; $d_\perp\to 2$ in the azimuthal term, which multiplies no pole).

6 Result

$$\frac{d^3N_{\rm NLO}}{d^3k} = \underbrace{(8)}_{\text{JIMWLK}} + \underbrace{(11)}_{\text{BFKL}} + \underbrace{\text{DGLAP (non-soft + soft pair)} + \text{RC}}_{(13)} + \underbrace{\text{endpoint (9)} + \text{boundary} + \text{azimuthal}}_{\text{finite terms}}\,, \tag{14}$$

with the boundary term $-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}(1-\frac{k^+}{\vee})\,\mathcal{I}$. This coincides term by term with the categories of the corrected Row-IV note; the difference is only the order of operations:

(i) both poles are removed *before* any dimensional regularization, so the two rapidity logarithms are produced directly by the two counterterm integrals — $\log\frac{k^+}{\Lambda}$ with the (z-form) LO $\otimes$ soft-kernel coefficient (JIMWLK) and $\log\frac{\vee}{k^+}$ with the UV-finite, μ-free production kernel (BFKL) — and neither ever multiplies L_ϵ;

(ii) the subtraction of T_B automatically leaves the soft channel in its μ-free paired form inside DGLAP: no term is ever counted twice and no term has to be moved between categories afterwards;

(iii) the only ultraviolet logarithms in the final result are the collinear L_ϵ of the non-soft DGLAP kernel, the RC term, and the endpoint term; the finite terms (endpoint, boundary, azimuthal) carry no large longitudinal logarithm.